BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: PR–IX (A-I-A4) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** PR–IX (A-I-A4) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The number of protons in chlorine atom is:

(A) 12 (B) 6

(C) 17 (D) 11

Q2. The relative atomic mass of oxygen atom is:

(A) 15.998 amu (B) 16.998 amu

(C) 14.998 amu (D) 17.998 amu

Q3. The f-subshell can accommodate a maximum of:

(A) 14 electrons (B) 10 electrons

(C) 06 electrons (D) 02 electrons

Q4. Which group of elements contains incomplete s-subshell?

(A) Group I-A (B) Group II-A

(C) Group III-A (D) Group IV-A

Q5. Helium obeys duplet rule. How many electrons are in its valance shell:

(A) 1 (B) 2

(C) 3 (D) 4

Q6. Which of the given two elements will form ionic bond;

(A) Na and K
(B) K and Ca
(C) Ca and Cl
(D) Cl and C

Q7. Mobility of particles in solid, liquid and gas depends on:

(A) Kinetic energy
(B) Diffusion
(C) Density
(D) Volume

Q8. Substance present in larger amount in a solution is called;

(A) Solute
(B) Salt
(C) Solution
(D) Solvent

Q9. Particle of which settle down at bottom on standing for some time:

(A) Dispersion
(B) Solution
(C) Colloids
(D) Suspension

Q10. Which one of the given acts as oxidizing agent in the given reaction? $Br_2 + H_2S \rightarrow 2HBr + S$

(A) Br_2
(B) H
(C) S
(D) H_2S

Q11. Reduction of alkali metals ion take place at:

(A) Anode
(B) Walls of cell
(C) Battery
(D) Cathode

Q12. Elements above calcium in IIA of periodic table with symbol and atomic number is:

(A) Na=11
(B) Mg=12
(C) C=13
(D) N=14

SECTION-B

Time Allowed: 2 Hours 45 Minutes

> Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 45 minutes.

Q2. Attempt any EIGHT parts. Each part carries FOUR marks.

1) Differentiate between physical chemistry and analytical chemistry.

2) Predict the number of proton and neutrons in148C and3177Cl.

3) Write factors which affect the ionization energies of atoms?

4) How many valance electrons are present in the following elements? (a) Barium (b) Carbon (c) Magnesium (d) Krypton

5) Draw diagram to show the ionic bond in MgO?

6) A gas occupies 4.31 liters at a pressure of 0.755 atm. Determine the volume if the pressure is increased to 1.25 atm.

7) Identify the solute and solvent, alcohol in water and carbonated drink.

8) Calculate the molarity of a solution containing 0.5 moles of HCl in 1.5 dm3.

9) Determine the oxidation number of hydrogen in the following compound. (a) HCl (b) NaH

10) Explain cell reaction in voltaic cell. Write half-cell reactions, and overall reaction.

11) Interpret the chemical reaction between: (i) Potassium and iodine (b) Hydrogen and iodine

SECTION-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 24

Note: Note: Attempt any THREE questions. All questions carry equal marks.

Q3.

(a) Calculate gram molecular mass of H2SO4.

(b) What do you know about Auf-Bau's principle? How many electrons can accommodate in M shell of an element?

Q4.

(a) In how many categories 7th period elements are classified?

(b) How ionic bond is formed between metal atom and non-metals atom. Explain with the help of example.

Q5.

(a) Explain solid state with reference to attractive forces, motion and rigidity.

(b) Complete the following table? Class of Colloid Phase Example Gell?? ?? Butter

Q6.

(a) Explain how Alloying and cathodic protection helps to prevent corrosion.

(b) Explain the formation of cations from the given elements Na, Ca and Al?